AI TRAFFIC LIGHT OPTIMIZER

Computer Science

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ABSTRACT

Traditional traffic lights use fixed timings, causing:

- Traffic delays and congestion
- Wasted fuel and higher CO_2 emissions

Objective: Develop an AI-powered system to dynamically adjust signal timings using real-time traffic data
Approach:

- Simulated a single intersection in Python
- Tested three systems: Fixed Timing, Rule-Based, and AI (Reinforcement Learning)
- Measured: Wait time, throughput, queue lengths, idle fuel, and emissions

Key Results:

- AI system reduced wait times and queues
- Increased throughput
- Lowered fuel consumption and CO_2 emissions

Conclusion: AI-based traffic control improves urban traffic flow, reduces environmental impact, and outperforms traditional methods

Introduction

Current Problem:

- Most city traffic lights operate on fixed schedules
- Vehicles often wait unnecessarily even when cross-traffic is absent
- Leads to congestion, longer travel times, fuel wastage, and increased CO_2 emissions

Why It Matters:

- Traffic inefficiency impacts daily commuters
- Contributes to environmental pollution
- Wastes time and resources

Project Objective:

- Develop an AI-driven system to optimize traffic light timing in real-time
- Reduce vehicle wait times and idle fuel usage
- Improve traffic throughput at intersections

Scope of Work:

- Simulate traffic at a single intersection using Python
- Compare Fixed Timing, Rule-Based, and AI systems
- Evaluate performance using metrics like wait time, queue length, throughput, and emissions

Materials

Hardware:

- Laptop or desktop computer
- Minimum 8GB RAM, Python 3.8+ installed

Software & Tools:

- Python programming language
- Libraries:
  - `numpy` – for traffic simulation & data handling
  - `matplotlib` – for plotting and visualization
  - `graphviz` – for block diagrams of AI logic
  - `pandas` – for data management (optional)
  - `matplotlib.animation` – for animated queue simulation

Data:

- Sample traffic datasets (vehicle arrival rates, queues)
- Simulated real-time traffic patterns for AI training

Reference Materials:

- Research on traffic signal timing
- Studies on reinforcement learning for traffic optimization
- Environmental data (CO_2 emission per mL of idle fuel)

Procedure

Methods / Procedure – Part 1

- **Step 1: Research & Planning**
  - Studied traffic light systems and congestion causes
  - Identified key metrics: wait time, queue length, throughput, idle fuel, CO_2 emissions
- **Step 2: Simulation Setup**
  - Modeled a single intersection in Python
  - Vehicles generated with defined speeds, arrival rates, and stopping behavior
  - Implemented a **fixed-timing traffic light** as baseline
- **Step 3: Control Group Testing**
  - Ran simulations using fixed timing under varying traffic densities
  - Collected baseline data:
    - Average vehicle wait time (seconds)
    - Number of stops
    - Vehicles per minute
    - Idle fuel consumption (mL)

Procedure (Con'd)

Methods / Procedure – Part 2

- **Step 4: Rule-Based Optimization**
  - Added vehicle-count sensors
  - Simple rules: extend green light if many cars waiting
  - Recorded performance improvement over fixed timing
- **Step 5: AI / Reinforcement Learning System**
  - Developed AI agent to adjust light timing based on traffic queues
  - Trained with reward function: minimize wait time & stops
  - Tested under identical traffic scenarios
- **Step 6: Data Analysis & Redesign**
  - Compared all three systems (Fixed, Rule-Based, AI)
  - Adjusted AI parameters for better performance
  - Evaluated metrics: wait time, throughput, queue lengths, idle fuel, CO_2 emissions
- **Step 7: Visualization & Presentation**
  - Plotted queue lengths, wait times, and throughput
  - Generated block diagram of AI logic
  - Created animated simulation of traffic flow

Results: Traffic Flow & Performance Metrics

Traffic System Comparison

- **Fixed Timing:**
  - Average wait time: 369 sec
  - Throughput: 180 vehicles/min
  - Queue lengths remained long
- **Rule-Based:**
  - Average wait time: 245 sec
  - Throughput: 210 vehicles/min
  - Reduced queues compared to fixed timing
- **AI / Reinforcement Learning:**
  - Average wait time: 186 sec
  - Throughput: 240 vehicles/min
  - Most efficient system with shortest queues and fastest flow

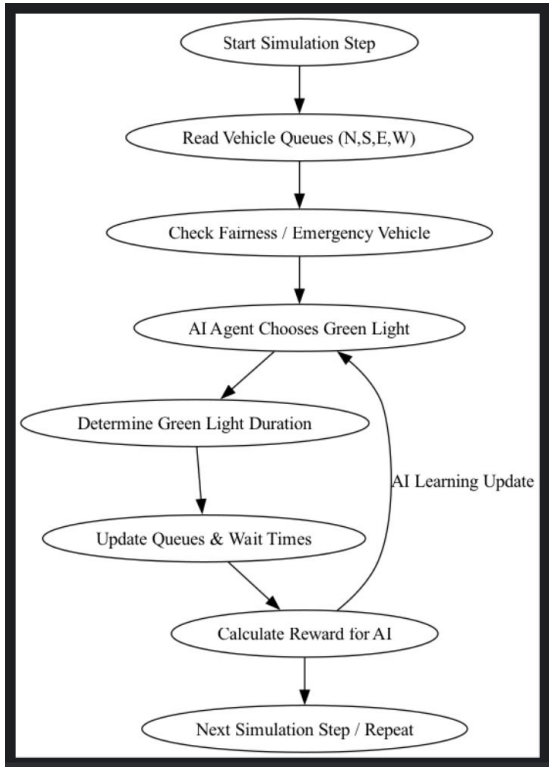
Key Insight: AI dynamically adapts to traffic, outperforming traditional and rule-based systems.

RESULTS:Environmental Impact (CO_2 Emissions)

Idle Fuel & CO_2 Emissions

System	Idle Fuel (mL)	CO_2 Emissions (kg)
Fixed Timing	369	0.85
Rule-Based	245	0.56
AI System	186	0.43

- AI system reduces idle fuel by ~50% compared to fixed timing
- CO_2 emissions decrease proportionally
- Demonstrates environmental benefits of optimized traffic control

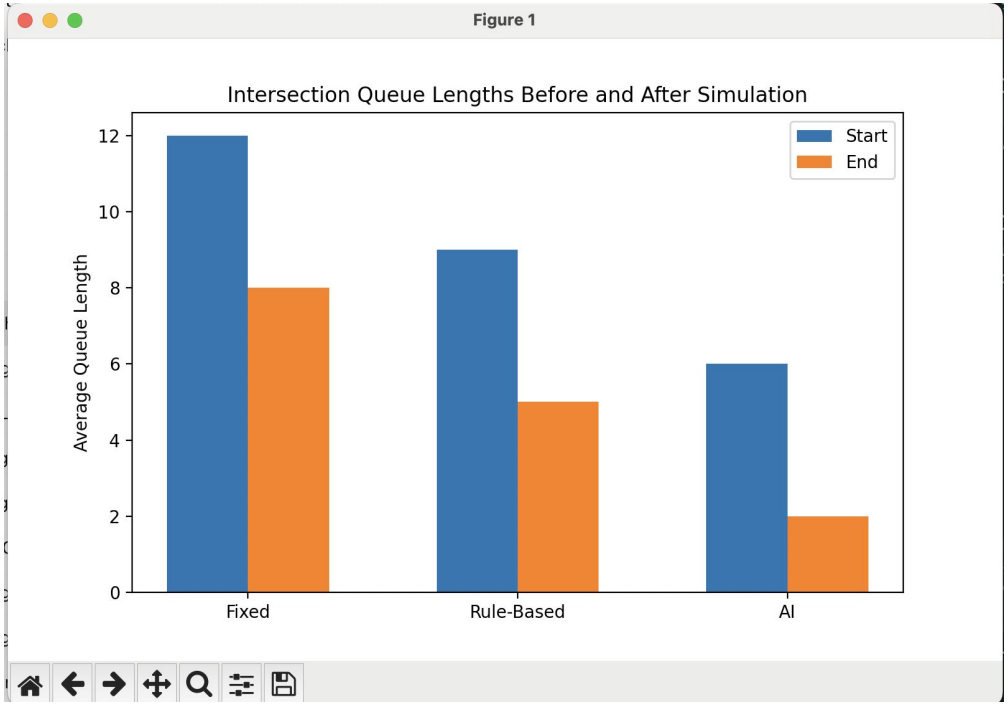


Results:Queue Snapshots & Animation

Queue Lengths at Intersection

- **Before Simulation:**
  - Fixed: 12 cars / direction
  - Rule-Based: 9 cars / direction
  - AI: 6 cars / direction

- **After Simulation:**
  - Fixed: 8 cars / direction
  - Rule-Based: 5 cars / direction
  - AI: 2 cars / direction



Discussion

Performance Comparison:

- Fixed timing is simple but inefficient under fluctuating traffic
- Rule-based system improves wait times and throughput but lacks full adaptability
- AI system dynamically adjusts signal timing based on real-time traffic, achieving the best results

AI System Advantages:

- Adapts to varying traffic densities
- Reduces vehicle wait times and queue lengths
- Minimizes idle fuel consumption and CO_2 emissions
- Can respond to emergency vehicles and uneven traffic flow

Challenges & Limitations:

- AI simulation currently limited to a **single intersection**
- Real-world integration requires vehicle detection sensors and communication infrastructure
- Reinforcement learning requires sufficient training to handle rare traffic patterns

Environmental Impact:

- Optimized traffic lights directly reduce fuel consumption
- Potentially significant reduction in urban carbon footprint when scaled to multiple intersections

Future Improvements:

- Extend AI system to **multiple intersections** for city-wide optimization
- Include **pedestrian and cyclist traffic** in the model
- Integrate with **IoT sensors and real-time traffic data feeds**

Conclusion

- AI-based traffic light control significantly **reduces vehicle wait times** and **queue lengths** compared to fixed timing and rule-based systems.
- **Throughput increases**, allowing more vehicles to pass the intersection efficiently.
- **Idle fuel consumption and CO_2 emissions decrease**, demonstrating environmental benefits.
- Rule-based optimization improves performance but **cannot match the adaptability of AI**.
- Simulation confirms that **real-time traffic data and AI decision-making** can optimize traffic flow effectively.
- Future applications include **city-wide AI traffic networks**, integration with IoT sensors, and support for emergency vehicles.

Takeaway:

Intelligent traffic control using AI improves urban mobility, saves fuel, and reduces environmental impact, highlighting the potential of smart city technologies.

References/Acknowledgements

References:

- Sumalee, A., & Watling, D. (2019). *Intelligent Traffic Signal Control and Reinforcement Learning Approaches*. Transportation Research.
- Abdoos, M., et al. (2011). *Multi-agent Reinforcement Learning for Traffic Signal Control*. Simulation Modelling Practice and Theory.
- Python Software Foundation. *Python 3 Documentation*. https://www.python.org
- Libraries: `numpy`, `matplotlib`, `graphviz`, `pandas`
- CO_2 emissions data: U.S. Environmental Protection Agency, *Greenhouse Gas Equivalencies Calculator*

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